

CS 2601 Linear and Convex Optimization

1. Introduction

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Outline

- Examples of optimization problems
- Global and local optima

Optimization problems

$$\begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{x} \in X \end{array} \quad \text{or} \quad \min_{\mathbf{x} \in X} f(\mathbf{x})$$

- $f : \mathbb{R}^n \rightarrow \mathbb{R}$: **objective function**
- $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$: **optimization/decision variables**
- $X \subset \mathbb{R}^n$: **feasible set** or **constraint set**
 - ▶ $\mathbf{x}$ is called **feasible** if $\mathbf{x} \in X$ and **infeasible** if $\mathbf{x} \notin X$.

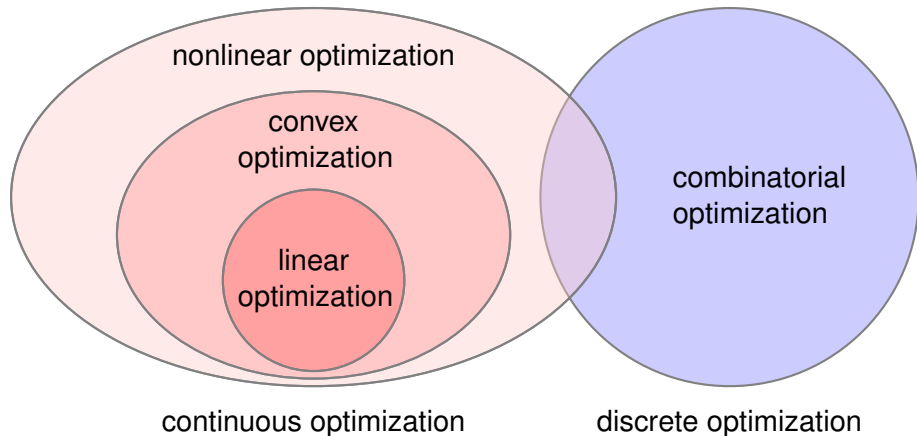
Maximizing f is equivalent to minimizing $-f$; will focus on minimization.

The problem is **unconstrained** if $X = \mathbb{R}^n$ and **constrained** if $X \neq \mathbb{R}^n$.

X is often specified by **constraint functions**,

$$\begin{array}{ll} \min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s. t.} & g_i(\mathbf{x}) \leq 0, \quad i = 1, 2, \dots, m \\ & h_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, k \end{array}$$

Optimization problems



Data fitting

Recall Hooke's law in physics,

$$F = -k(x - x_0) = -kx + b, \quad \text{where } b = kx_0$$

- F : force
- k : spring constant
- x : length
- x_0 : length at rest

Given m measurements $(x_1, F_1), (x_2, F_2), \dots, (x_m, F_m)$,

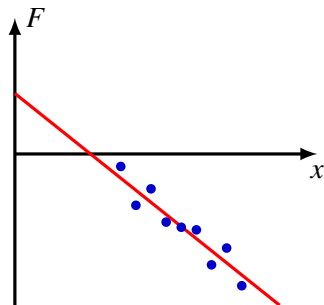
$$F_i = -kx_i + b + \epsilon_i$$

- ϵ_i : measurement error

find k, b by fitting a line through data.

Least squares criterion,

$$\min_{k>0, b>0} \sum_{i=1}^m \epsilon_i^2 = \sum_{i=1}^m (F_i + kx_i - b)^2$$



Linear least squares regression

A **linear model** predicts a **response/target** by a linear combination of **predictors/features** (plus an **intercept/bias**),

$$\hat{y} = f(\mathbf{x}) = b + \sum_{i=1}^n w_i x_i = \mathbf{x}^T \mathbf{w} + b$$

Given m data points $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)$, **linear (least squares) regression** finds $\mathbf{w}$ and b by minimizing the sum of squared errors,

$$\min_{\mathbf{w} \in \mathbb{R}^n, b \in \mathbb{R}} \sum_{i=1}^m (f(\mathbf{x}_i) - y_i)^2 = \sum_{i=1}^m (\mathbf{x}_i^T \mathbf{w} + b - y_i)^2$$

In a more compact form,

$$\min_{\mathbf{w} \in \mathbb{R}^n, b \in \mathbb{R}} \|\mathbf{X}\mathbf{w} + b\mathbf{1} - \mathbf{y}\|^2$$

- $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_m]^T \in \mathbb{R}^{m \times n}, \mathbf{y} = (y_1, \dots, y_m) \in \mathbb{R}^m$
- $\mathbf{1} = (1, 1, \dots, 1) \in \mathbb{R}^m$
- $\|\mathbf{z}\| = \sqrt{\mathbf{z}^T \mathbf{z}} = \sqrt{\sum_{i=1}^n z_i^2}$ for $\mathbf{z} = (z_1, \dots, z_n) \in \mathbb{R}^n$

Optimal transport problem

- need to ship products from n warehouses to m customers
- inventory at warehouse i is a_i , $i = 1, 2, \dots, n$
- quantity ordered by customer j is b_j , $j = 1, 2, \dots, m$
- unit shipping cost from warehouse i to customer j is c_{ij}

Let x_{ij} be quantity shipped from warehouse i to customer j

Minimize total cost by solving the following **linear program**

$$\begin{aligned} \min_{(x_{ij})} \quad & \sum_{i=1}^n \sum_{j=1}^m c_{ij} x_{ij} \\ \text{s. t.} \quad & \sum_{i=1}^n x_{ij} = b_j \quad \text{for } j = 1, 2, \dots, m \\ & \sum_{j=1}^m x_{ij} \leq a_i \quad \text{for } i = 1, 2, \dots, n \\ & x_{ij} \geq 0 \quad \text{for } i = 1, 2, \dots, n; j = 1, 2, \dots, m \end{aligned}$$

Power allocation

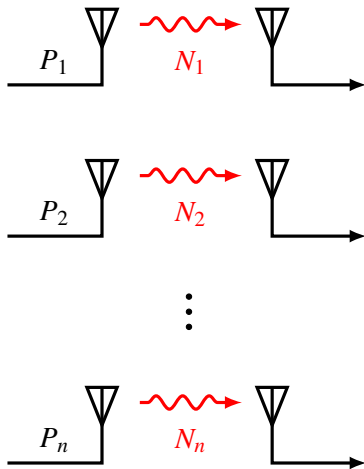
We want to transmit information over n communication channels. The capacity of the i -th channel is

$$C_i = W_i \log_2 \left(1 + \frac{P_i}{N_i} \right) \quad \text{bits/second}$$

- W_i : channel bandwidth in hertz
- P_i : signal power in watts
- N_i : noise power in watts

Given a total power constraint P ,

$$\begin{aligned} \max_{P_1, \dots, P_n} \quad & \sum_{i=1}^n W_i \log_2 \left(1 + \frac{P_i}{N_i} \right) \\ \text{s. t.} \quad & \sum_{i=1}^n P_i \leq P, \quad P_i \geq 0 \quad \text{for } i = 1, 2, \dots, n \end{aligned}$$



Binary classification



vs



Represent an image by a vector $\mathbf{x} \in \mathbb{R}^n$, label $y \in \{+1, -1\}$

Given a set of images with labels $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)$, want a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, called a **classifier**, such that

$$\begin{cases} f(\mathbf{x}_i) > 0, & \text{iff } y_i = +1 \\ f(\mathbf{x}_i) < 0, & \text{iff } y_i = -1 \end{cases} \iff y_i f(\mathbf{x}_i) > 0$$

Once we find f , we can use $\hat{y} = \text{sign}[f(\mathbf{x})]$ to classify new images.

How to find f ? Let's consider **linear classifiers**, i.e. $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$, and two methods to determine $\mathbf{w}, b$, i.e. **logistic regression** and **support vector machine**.

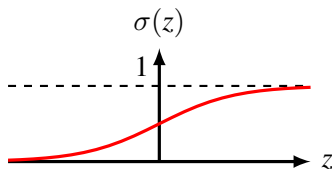
Logistic regression

Logistic regression assumes the probability for an image x to have label y is modeled by

$$p(y | x) = \sigma(yf(x)) = \sigma(y(\mathbf{w}^T \mathbf{x} + b))$$

where σ is the sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



To determine $\mathbf{w}, b$, maximize the likelihood

$$\max_{\mathbf{w}, b} L(\mathbf{w}, b) = \prod_{i=1}^m p(y_i | x_i)$$

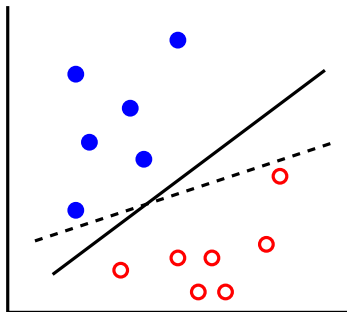
or equivalently, minimize the negative log likelihood

$$\min_{\mathbf{w}, b} NLL(\mathbf{w}, b) = -\log L(\mathbf{w}, b) = \sum_{i=1}^m \log(1 + e^{-y_i(\mathbf{x}_i^T \mathbf{w} + b)})$$

Support vector machine

Assume data is linearly separable, i.e. exists hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$
s.t.

$$y_i(\mathbf{w}^T \mathbf{x}_i + b) > 0, \quad \forall i$$



There may exist many such hyperplanes with different **margins**, i.e. minimum distance from data points to the hyperplane.

- A classifier with a larger margin is more robust against noise.

Support vector machine

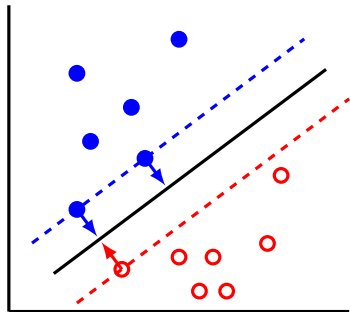
Given a hyperplane $P : \mathbf{w}^T \mathbf{x} + b = 0$,

- distance from $\mathbf{x}_i$ to P ,

$$\text{dist}(\mathbf{x}_i, P) = \frac{|\mathbf{w}^T \mathbf{x}_i + b|}{\|\mathbf{w}\|}$$

- margin

$$\min_{1 \leq i \leq m} \frac{|\mathbf{w}^T \mathbf{x}_i + b|}{\|\mathbf{w}\|}$$



Support vector machine (SVM) maximizes the margin

$$\begin{aligned} \max_{\mathbf{w}, b} \quad & \min_{1 \leq i \leq m} \frac{|\mathbf{w}^T \mathbf{x}_i + b|}{\|\mathbf{w}\|} \\ \text{s. t.} \quad & y_i(\mathbf{w}^T \mathbf{x}_i + b) > 0, \quad i = 1, 2, \dots, m \end{aligned}$$

Not easy to solve in this form.

Support vector machine

Problem reformulation

- Note $|\mathbf{w}^T \mathbf{x}_i + b| = y_i(\mathbf{w}^T \mathbf{x}_i + b)$, as $y_i = \text{sign}(\mathbf{w}^T \mathbf{x}_i + b)$.
- For $\alpha > 0$, $\tilde{\mathbf{w}} = \alpha \mathbf{w}$ and $\tilde{b} = \alpha b$ determine the same hyperplane P ,

$$\mathbf{x} \in P \iff \mathbf{w}^T \mathbf{x} + b = 0 \iff \tilde{\mathbf{w}}^T \mathbf{x} + \tilde{b} = 0$$

- Choosing α properly, we can assume $\min_{1 \leq i \leq m} y_i(\tilde{\mathbf{w}}^T \mathbf{x}_i + \tilde{b}) = 1$,

$$\begin{aligned} \max_{\tilde{\mathbf{w}}, \tilde{b}} \quad & \frac{1}{\|\tilde{\mathbf{w}}\|} \\ \text{s. t.} \quad & y_i(\tilde{\mathbf{w}}^T \mathbf{x}_i + \tilde{b}) \geq 1, \quad i = 1, 2, \dots, m \end{aligned}$$

- For $z > 0$, maximizing $\frac{1}{z}$ is equivalent to minimizing $\frac{1}{2}z^2$,

$$\begin{aligned} \min_{\tilde{\mathbf{w}}, \tilde{b}} \quad & \frac{1}{2} \|\tilde{\mathbf{w}}\|^2 \\ \text{s. t.} \quad & y_i(\tilde{\mathbf{w}}^T \mathbf{x}_i + \tilde{b}) \geq 1, \quad i = 1, 2, \dots, m \end{aligned}$$

We will see this reformulation is a convex problem and easy to solve. 12

Appendix: Distance to hyperplane

- $\mathbf{w} \perp$ hyperplane $P : \mathbf{w}^T \mathbf{x} + b = 0$
- $\mathbf{x}'_i$ is the **orthogonal projection** of $\mathbf{x}_i$ onto P , i.e.

$$\mathbf{x}_i - \mathbf{x}'_i \perp P$$

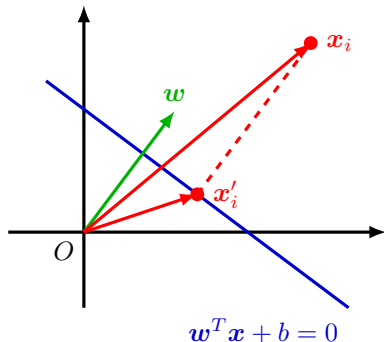
$$\mathbf{w}^T \mathbf{x}'_i + b = 0$$

- $\mathbf{x}_i - \mathbf{x}'_i = \gamma_i \mathbf{w}$ for some $\gamma_i \in \mathbb{R}$,

$$\mathbf{w}^T (\mathbf{x}_i - \gamma_i \mathbf{w}) + b = 0 \implies \gamma_i = \frac{\mathbf{w}^T \mathbf{x}_i + b}{\mathbf{w}^T \mathbf{w}}$$

- distance from $\mathbf{x}_i$ to P is

$$\min_{\mathbf{x} \in P} \|\mathbf{x}_i - \mathbf{x}\| = \|\mathbf{x}_i - \mathbf{x}'_i\| = \|\gamma_i \mathbf{w}\| = \frac{|\mathbf{w}^T \mathbf{x}_i + b|}{\|\mathbf{w}\|}$$



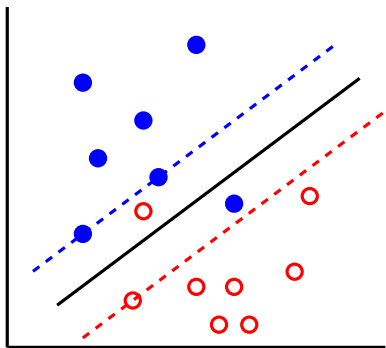
Soft margin SVM

Hard margin SVM requires linear separability

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s. t.} \quad & y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \quad \forall i \end{aligned}$$

When not linearly separable,

- relax constraints
- penalize deviation



Soft margin SVM: introduce **slack variables** $\xi = (\xi_1, \dots, \xi_n)^T$

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{i=1}^m \xi_i \quad (C > 0 \text{ is a hyperparameter}) \\ \text{s. t.} \quad & y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i, \quad i = 1, 2, \dots, m \\ & \xi \geq \mathbf{0}, \quad (\text{i.e.} \quad \xi_i \geq 0, \quad i = 1, 2, \dots, m) \end{aligned}$$

Outline

- Examples of optimization problems
- Global and local optima

Global optima

$\mathbf{x}^* \in X$ is a **global minimum**¹ of f over X if

$$f(\mathbf{x}^*) \leq f(\mathbf{x}), \quad \forall \mathbf{x} \in X$$

It is an **optimal solution** of the minimization problem

$$\min_{\mathbf{x} \in X} f(\mathbf{x}) \quad (\text{P})$$

and $f(\mathbf{x}^*)$ is the **optimal value** of (P).

Global maximum is defined by reversing the direction of the inequality.

Maximum and minimum are called **extremum**.

Note. Global extrema may not exist.

- $f(x) = x$, $X = \mathbb{R}$, $\inf_{x \in X} f(x) = -\infty$ unbounded below
- $f(x) = x$, $X = (0, 1)$, $\inf_{x \in X} f(x) = 0$, but not achievable

¹**Global minimum** often also refers to the minimum value $f(\mathbf{x}^*)$.

Review: Inner product and norm

Euclidean inner product on $\mathbb{R}^n$: $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$

Euclidean norm (2-norm, ℓ_2 -norm): $\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^T \mathbf{x}} = \sqrt{\sum_{i=1}^n x_i^2}$

A **norm** on $\mathbb{R}^n$ is a function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfying

1. $\|\mathbf{x}\| \geq 0, \forall \mathbf{x} \in \mathbb{R}^n$
 2. $\|\mathbf{x}\| = 0$ iff $\mathbf{x} = \mathbf{0}$
 3. $\|a\mathbf{x}\| = |a|\|\mathbf{x}\|, \forall a \in \mathbb{R}, \mathbf{x} \in \mathbb{R}^n$
 4. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$
- (positive definiteness) (positive homogeneity) (triangle inequality)

Example.

- p -norm (ℓ_p -norm): $\|\mathbf{x}\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}, p \geq 1$
- ∞ -norm (ℓ_∞ -norm): $\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i| = \lim_{p \rightarrow \infty} \|\mathbf{x}\|_p$

Property 4 of these norms is given by Minkowski's inequality.

By default, $\|\mathbf{x}\|$ means $\|\mathbf{x}\|_2$.

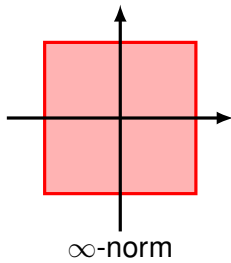
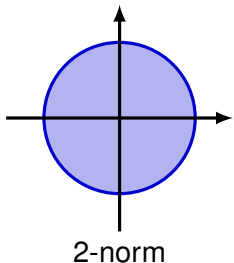
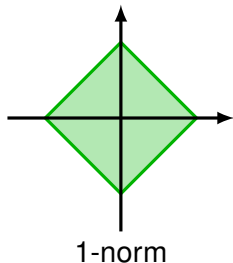
Review: Open and closed balls

Open ball of radius r centered at $\mathbf{x}_0$

$$B(\mathbf{x}_0, r) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_0\| < r\}$$

Closed ball of radius r centered at $\mathbf{x}_0$

$$\bar{B}(\mathbf{x}_0, r) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_0\| \leq r\}$$



unit balls in $\mathbb{R}^2$ with different norms

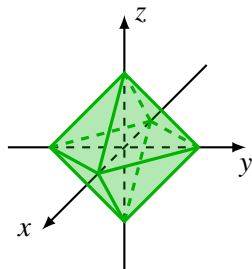
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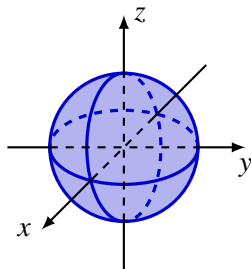
$$B(\mathbf{x}_0, r) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_0\| < r\}$$

Closed ball of radius r centered at $\mathbf{x}_0$

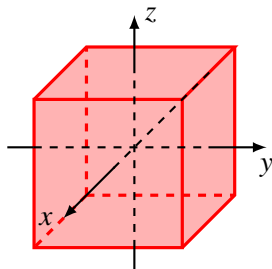
$$\bar{B}(\mathbf{x}_0, r) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_0\| \leq r\}$$



1-norm



2-norm



∞ -norm

unit balls in $\mathbb{R}^3$ with different norms

Review: Open and closed sets

A set S is **open** if for any $x \in S$, there exists $\epsilon > 0$ s.t. $B(x, \epsilon) \subset S$.

A set S is **closed** if its complement S^c is open.

Examples in $\mathbb{R}$.

- $(0, 1)$ is open.
- $[0, 1]$ is closed.
- $(0, 1]$ is neither open nor closed.
- $[1, \infty)$ is closed.

A sequence $\{x_n\}$ converges to x , denoted $x_n \rightarrow x$ or $\lim_{n \rightarrow \infty} x_n = x$ if

$$\lim_{n \rightarrow \infty} \|x - x_n\| = 0$$

Note. In $\mathbb{R}^n$, if $x_n \rightarrow x$ in one norm, it converges in any norm. The convergence in norm is equivalent to coordinate-wise convergence.

Theorem. S is closed iff for any sequence $\{x_n\} \subset S$,

$$x_n \rightarrow x \implies x \in S.$$

Review: Compactness

A set S is **bounded** if there exists $M < \infty$ s.t. $\|\mathbf{x}\| \leq M, \forall \mathbf{x} \in S$.

A set $S \subset \mathbb{R}^n$ is **compact** if it is closed and bounded.

Examples in $\mathbb{R}$.

- $[0, 1]$ is compact
- $(0, 1), (0, 1]$ are bounded but not closed, so not compact
- $[1, \infty)$ is closed but not bounded, so not compact

Examples in $\mathbb{R}^n$.

- For $0 < r < \infty$, the open ball $B(0, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| < r\}$ is not compact.
- For $r < \infty$, the closed ball $\bar{B}(0, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \leq r\}$ is compact.
- $\{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \geq \mathbf{0}\}$ is closed but unbounded, so not compact.

Review: Continuity

A function $f : X \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is **continuous at $x \in X$** if for any $\epsilon > 0$, there exists $\delta > 0$ s.t.

$$\mathbf{y} \in X \cap B(\mathbf{x}, \delta) \implies |f(\mathbf{y}) - f(\mathbf{x})| < \epsilon$$

Equivalently, f is continuous at $\mathbf{x} \in X$ if

$$\forall \{\mathbf{x}_n\} \subset X, \quad \mathbf{x}_n \rightarrow \mathbf{x} \implies f(\mathbf{x}_n) \rightarrow f(\mathbf{x})$$

f is **continuous** on X if it is continuous at every $\mathbf{x} \in X$.

Theorem. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous on $\mathbb{R}^n$, then for any $c \in \mathbb{R}$,

1. $\{\mathbf{x} : f(\mathbf{x}) < c\}$ is open;
2. $\{\mathbf{x} : f(\mathbf{x}) \leq c\}$ is closed.

Existence of Global Optima

Extreme Value Theorem. If f is continuous on a compact set X , then f attains its maximum and minimum on X , i.e. there exist $x_1, x_2 \in X$ (not necessarily unique) s.t.

$$f(x_1) \leq f(x) \leq f(x_2), \quad \forall x \in X.$$

Example. $f(x) = x^2$ satisfies $f(0) \leq f(x) \leq f(2) = f(-2)$ on $[-2, 2]$.

The Extreme Value Theorem gives **sufficient** conditions for the existence of global optima, but they are **not necessary**.

Example. $f(x) = x^2$.

- $\inf_{x \in (0,1)} f(x) = 0$, but $f(x) > 0$ for all $x \in (0, 1)$, no global min.
- $\min_{x \in [0,1)} f(x) = f(0)$, $x^* = 0$ is global min, but $[0, 1)$ is not closed.
- $\min_{x \in \mathbb{R}} f(x) = f(0)$, $x^* = 0$ is global min, but $\mathbb{R}$ unbounded.

Existence of Global Optima (cont'd)

Corollary. If f is continuous on $\mathbb{R}^n$ and $f(\mathbf{x}) \rightarrow +\infty$ as $\|\mathbf{x}\| \rightarrow \infty$, then the global min exists, i.e. there exists $\mathbf{x}^*$ s.t. $f(\mathbf{x}^*) \leq f(\mathbf{x})$, $\forall \mathbf{x}$.

Proof.

- Since $f(\mathbf{x}) \rightarrow +\infty$ as $\|\mathbf{x}\| \rightarrow \infty$, there exists $M > 0$ s.t. $f(\mathbf{x}) > f(\mathbf{0})$ when $\|\mathbf{x}\| > M$.
- The closed ball $\bar{B}(\mathbf{0}, M) = \{\mathbf{x} : \|\mathbf{x}\| \leq M\}$ is compact.
- By the Extreme Value Theorem, there exists $\mathbf{x}^* \in \bar{B}(\mathbf{0}, M)$ s.t.

$$f(\mathbf{x}^*) \leq f(\mathbf{x}), \quad \forall \mathbf{x} \in \bar{B}(\mathbf{0}, M)$$

- For $\mathbf{x} \notin \bar{B}(\mathbf{0}, M)$, $f(\mathbf{x}^*) \leq f(\mathbf{0}) < f(\mathbf{x})$, so $\mathbf{x}^*$ is a global min on $\mathbb{R}^n$.

A function f is called **coercive** if $f(\mathbf{x}) \rightarrow +\infty$ as $\|\mathbf{x}\| \rightarrow \infty$.

Example. $f(\mathbf{x}) = \|\mathbf{x}\|^2$ coercive, $\mathbf{x}^* = \mathbf{0}$ is global minimum.

Example. $f(\mathbf{x}) = e^{-\|\mathbf{x}\|}$ not coercive, no global minimum.

Example. $f(x) = \sin x$ not coercive, $x^* = -\frac{\pi}{2}$ is global minimum.

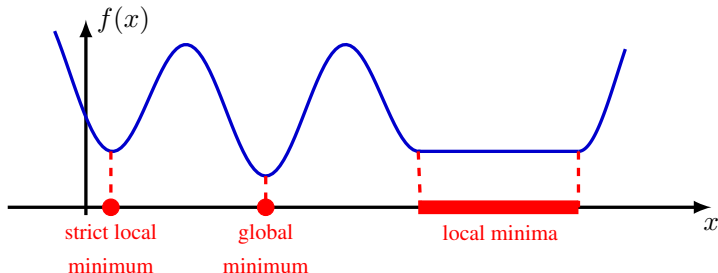
Local Minimum

$\mathbf{x}^* \in X$ is a **local minimum** of f on X if there exists $\epsilon > 0$ s.t.

$$f(\mathbf{x}^*) \leq f(\mathbf{x}), \quad \forall \mathbf{x} \in X \cap B(\mathbf{x}^*, \epsilon)$$

$\mathbf{x}^*$ is a **strict local minimum** if strict inequality holds for $\mathbf{x} \neq \mathbf{x}^*$.

Local maximum is defined by reversing direction of inequality.



Global minimum is always local minimum, but **not** vice versa.

- We will see local min is global min for convex problems

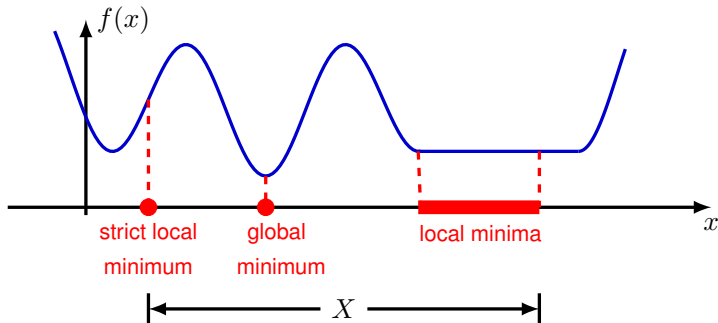
Local Minimum

$\mathbf{x}^* \in X$ is a **local minimum** of f on X if there exists $\epsilon > 0$ s.t.

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